

```
DROP DATABASE IF EXISTS biblioteca_db;
```

```
CREATE DATABASE biblioteca_db;
```

```
USE biblioteca_db;
```

```
-- Seguridad y usuarios
```

```
CREATE TABLE person (
```

```
    id_person INT PRIMARY KEY AUTO_INCREMENT,
```

```
    full_name VARCHAR(100) NOT NULL,
```

```
    email VARCHAR(100),
```

```
    phone VARCHAR(20)
```

```
);
```

```
CREATE TABLE user (
```

```
    id_user INT PRIMARY KEY AUTO_INCREMENT,
```

```
    username VARCHAR(50) UNIQUE NOT NULL,
```

```
    password VARCHAR(255) NOT NULL,
```

```
    person_id INT NOT NULL,
```

```
    FOREIGN KEY (person_id) REFERENCES person(id_person)
```

```
);
```

```
CREATE TABLE role (
```

```
    id_role INT PRIMARY KEY AUTO_INCREMENT,
```

```
    name VARCHAR(50) NOT NULL,
```

```
    description TEXT
```

```
);
```

```
CREATE TABLE role_user (  
    id_role_user INT PRIMARY KEY AUTO_INCREMENT,  
    user_id INT NOT NULL,  
    role_id INT NOT NULL,  
    FOREIGN KEY (user_id) REFERENCES user(id_user),  
    FOREIGN KEY (role_id) REFERENCES role(id_role)  
);
```

```
CREATE TABLE module (  
    id_module INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(50) NOT NULL,  
    description TEXT  
);
```

```
CREATE TABLE module_role (  
    id_module_role INT PRIMARY KEY AUTO_INCREMENT,  
    module_id INT NOT NULL,  
    role_id INT NOT NULL,  
    FOREIGN KEY (module_id) REFERENCES module(id_module),  
    FOREIGN KEY (role_id) REFERENCES role(id_role)  
);
```

```
CREATE TABLE form (  
    id_form INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(50) NOT NULL,  
    url VARCHAR(255) NOT NULL
```

);

```
CREATE TABLE form_module (  
    id_form_module INT PRIMARY KEY AUTO_INCREMENT,  
    form_id INT NOT NULL,  
    module_id INT NOT NULL,  
    FOREIGN KEY (form_id) REFERENCES form(id_form),  
    FOREIGN KEY (module_id) REFERENCES module(id_module)  
);
```

-- Entidades de biblioteca

```
CREATE TABLE student (  
    id_student INT PRIMARY KEY AUTO_INCREMENT,  
    person_id INT NOT NULL,  
    grade VARCHAR(10),  
    enrollment_code VARCHAR(20),  
    FOREIGN KEY (person_id) REFERENCES person(id_person)  
);
```

```
CREATE TABLE book (  
    id_book INT PRIMARY KEY AUTO_INCREMENT,  
    title VARCHAR(150),  
    author VARCHAR(100),  
    isbn VARCHAR(20),  
    status VARCHAR(20) -- Disponible, Prestado, Atrasado  
);
```

```
CREATE TABLE category (  
    id_category INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(50),  
    description TEXT  
);
```

```
CREATE TABLE book_category (  
    id_book_category INT PRIMARY KEY AUTO_INCREMENT,  
    book_id INT NOT NULL,  
    category_id INT NOT NULL,  
    FOREIGN KEY (book_id) REFERENCES book(id_book),  
    FOREIGN KEY (category_id) REFERENCES category(id_category)  
);
```

```
CREATE TABLE loan (  
    id_loan INT PRIMARY KEY AUTO_INCREMENT,  
    book_id INT NOT NULL,  
    student_id INT NOT NULL,  
    loan_date DATE,  
    return_date DATE,  
    returned BOOLEAN DEFAULT FALSE,  
    FOREIGN KEY (book_id) REFERENCES book(id_book),  
    FOREIGN KEY (student_id) REFERENCES student(id_student)  
);
```

```
CREATE TABLE loan_history (  
    id_history INT PRIMARY KEY AUTO_INCREMENT,  
    loan_id INT NOT NULL,  
    registered_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    action VARCHAR(50),  
    FOREIGN KEY (loan_id) REFERENCES loan(id_loan)  
);
```

```
CREATE TABLE notification (  
    id_notification INT PRIMARY KEY AUTO_INCREMENT,  
    student_id INT NOT NULL,  
    message TEXT,  
    sent_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (student_id) REFERENCES student(id_student)  
);
```

```
CREATE TABLE system_config (  
    id_config INT PRIMARY KEY AUTO_INCREMENT,  
    config_key VARCHAR(50),  
    config_value TEXT,  
    description TEXT  
);
```